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Session 6: "Dakota Office Products" (HBS 9-102-021)

1. *Develop an activity-based cost system for Dakota Office Products (DOP) based on Year 2000 data. Feel free to use charts or diagrams to describe it. Calculate the activity cost driver rate for each DOP activity in 2000.*

Warehousing expenses	52	per unit
Freight	6	per delivery
Delivery Expenses	220	per delivery
Order Entry per hr	80	per hour
Manual Order Setup	10	per order
EDI Validation	5	per order
Manual Line Entry	4	per line
Receivable Interest	10%	Annual

Figure 1: Prorated Unit Cost

2. *Using your answer to Question 1, calculate the profitability of Customer A and Customer B.*

	A	B
Sales	103.000	104.000
Cost of Items Purchased	85.000	85.000
Gross margin	**18.000**	**19.000**
Warehousing expenses	10.400	10.400
Commercial Freight	1.200	900
Delivery	-	5.500
Total Order Cost (Manual + EDI)	330	1.720
Interest Cost	900	3.000
Total Cost	12.830	21.520
Profit	**5.170**	**(2.520)**
Current Profitability Method	5.250	6.250
Variance	2%	140%

Figure 2: Activity Based Accounting

3. *What explains any difference in profitability between the two customers? What are the limitations, if any, to the estimates of the profitability of the two customers?*

Although both customers have the same warehouse handling and storage (200 × $52) as well as interest rate (10% of average accounts receivable), they have significant differences. The difference in profitability between the two customors can be explained by:

- All 200 of A's cartons went by commercial freight; for B, 150 went by freight and 50 by desktop delivery, so B's freight is lower (150 × $6).
- All desktop deliveries (25 × $220) were for customer B.
- Customer A had a total of 6 manual orders at $10 each, 60 lines at $4 each as well as 6 EDI orders at $5 each. For a total of $330.
- Customer B, on the other hand, did not have EDI orders, with only 100 manual orders at $10 each and 180 lines at $4 each. For a total of $1,720.

- Although they have the same interest rate (10%), the interest cost is very different. Slow payment (90+ days vs about 30) leaves $30,000 of receivables outstanding vs $9,000, costing $3,000 vs $900, for B and A, respectively.
- Desktop deliveries cost $5,500 for B while customer A had 0.
- Customer B has 100 small manual orders for a cost of $1,720 vs $330 for customer A's 12 larger orders, half of them (6) placed by EDI.

The current method shows that customer B as the more profitable than customer B ($6,250 vs $5,250). Activity-based costing reverses this ($5.170 vs -$2.520).

Limitations

- Activity splits and time per task come from interviews and a few days of observation at one distribution center, so the estimates may be questionned.
- There is no clear way to allocate general and selling (G&S) costs between customers.
- The case treats every carton as costing the same to handle and ship, regardless of weight or distance.
- Desktop delivery is costed per delivery, not by distance or cartons delivered.

4. *Suppose that a major customer switched from placing all its orders manually to placing all its orders over the internet site. How should this affect the activity cost drivers' rates calculated in Question 1? How would the switch affect Dakota's profitability?*
5. *Assume that Dakota applies the analysis done in Question 2 to its entire customer base. How could such information help Dakota managers increase company profits?*